

BRIEF COMMUNICATIONS

On Four-Sheeted Polynomial Mappings of $\mathbb{C}^2$. The General Case

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Introduction

Let $\tilde{\mathbb{C}}^2$ and $\mathbb{C}^2$ be two copies of the complex plane, $f: \tilde{\mathbb{C}}^2 \rightarrow \mathbb{C}^2$ be a polynomial mapping whose Jacobian is a nonzero constant. The well-known Jacobian hypothesis asserts that such a mapping f is polynomially invertible (see surveys in [1, 2]). Our mapping f is polynomially invertible if and only if a generic point in $\mathbb{C}^2$ has exactly one preimage. The number of preimages of a generic point is called the *topological degree* of the mapping. It is known that polynomial mappings whose Jacobian is a nonzero constant cannot have topological degree 2 or 3 (see [3]). In this paper we consider the case of degree 4. The main result is the following theorem

Theorem. *If a polynomial mapping $f: \tilde{\mathbb{C}}^2 \rightarrow \mathbb{C}^2$ has topological degree 4, then its Jacobian is not a nonzero constant.*

This paper continues the research started in the paper [4], in which the theorem is proved for the case in which the mapping f has one dicritical component. Here we discuss the remaining cases.

Outline of the proof

Suppose, on the contrary, that there exists a polynomial mapping f of degree 4 whose Jacobian is a nonzero constant.

1. Dicritical components and their resolutions at infinity. By a finite number of σ -processes at infinity we can extend our mapping f to a regular rational mapping $F: \tilde{X} \rightarrow X$, where $\tilde{X}, X$ are nonsingular compact complex varieties containing, respectively, $\tilde{\mathbb{C}}^2$ and $\mathbb{C}^2$ as open subsets; the curves $\tilde{L} := \tilde{X} \setminus \tilde{\mathbb{C}}^2$ and $L := X \setminus \mathbb{C}^2$ consist of a finite number of rational components with transversal and at most pairwise intersections. As in [4], let $\tilde{L}_\infty := F^{-1}(L)$. Irreducible components of the curve $\tilde{L}$ will be marked by tildes; their F -images will be denoted by the same letters without tildes. For example, $a := F(\tilde{a})$. Irreducible components $\tilde{l} \subset \tilde{L} - \tilde{L}_\infty$ on which F is not constant are called *dicritical components* of the mapping F . Assume that the mapping F satisfies the minimality condition (see [4]). Then we can choose $\tilde{X}$ and X in such a way that the following propositions hold.

Proposition 1. *The mapping F has two dicritical components $\tilde{g}_1$ and $\tilde{g}_2$ and F is locally biholomorphic on $(\tilde{g}_1 \cup \tilde{g}_2) \setminus \tilde{L}_\infty$, has a ramification of order one along $\tilde{g}_1$ and a ramification of order two along $\tilde{g}_2$, $\tilde{L} = \tilde{L}_\infty + \tilde{g}_1 + \tilde{g}_2$. Each of the curves $\tilde{g}_1, \tilde{g}_2$ intersects $\tilde{L}_\infty$ at a single point.*

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Proposition 2. Each of the curves g_1, g_2 intersects L transversally at a single point. Possible splice-diagrams of the curve $g = g_1 \cup g_2$ near L are represented in Figs. 1-3. The right arrow in each figure corresponds to g_2 , the upper arrow to g_1 . For each nodal vertex $p \in L$ denote by R_p, L_p, D_p, A_p respectively the right, the left, the lower, and the upper branches of the graph L starting at p (possibly empty). Then we have

- 1) $\det L = -1$;
- 2) $\det A_p = 1$, $\det R_p > 1$ for all $p \in A_h$;
- 3) if $p \in L - A_h$, $p \neq h$ then $\det R_p = 1$, $\det D_p > 1$, $\det L_p > 1$;
- 4) in the case represented in Fig. 1 we have $\det A_h = 1$, $\det R_h = 1$, $\det D_h > 1$, $\det L_h > 1$;
- 5) in the case represented in Fig. 3 if $\det A_h \neq 1$, then $\det R_h = 1$, $\det L_h > 1$.

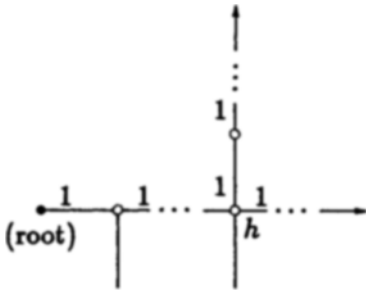


FIG. 1

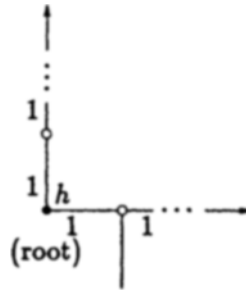


FIG. 2

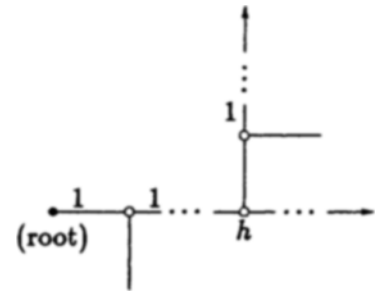


FIG. 3

In what follows, we use the notation from [4, §1]. As in [4, Lemma 6], the degree of the linear chain incident to $\tilde{g}_1$ is equal to 1, the degree of the linear chain incident to $\tilde{g}_2$ is equal to 2.

2. Structure of the graph $\tilde{L}_\infty$. In the paper [4], the graph $\tilde{L}_\infty$ was represented as the union of forks and maximal subgraphs of constant degree connecting them. Recall that a vertex $\tilde{l}$ is called a *fork* if $m(\tilde{l}) > 1$; a subgraph of constant degree is called *maximal* if it is not properly contained in another subgraph of constant degree. Here we introduce a different class of subgraphs connecting forks.

Definition 1. Suppose $a \in L$, A_h is the upper branch of L (see Proposition 2). Let us denote $a' := \{\tilde{a} \in F^{-1}(a) \text{ on which } F \text{ is not constant}\}$, $A' = F^{-1}(A_h)$. Let $\tilde{h} \in h'$. Suppose that either $m(\tilde{h}) = 1$ or $\det A_h = 1$ and L is such as in Fig. 2 or such as in Fig. 3.

Define a subgraph $\Delta_{\tilde{h}}$ as follows. If $m(\tilde{h}) \neq 1$, then $\Delta_{\tilde{h}}$ consists of all maximal subgraphs of constant degree incident to $\tilde{h}$ and all connected components of the set A' incident to $\tilde{h}$; if $m(\tilde{h}) = 1$, then $\Delta_{\tilde{h}}$ consists of the maximal subgraph of constant degree containing $\tilde{h}$ and connected components of the set A' incident to $\tilde{h}$.

The graph $\tilde{L}_\infty$ can be represented as the union of forks connecting subgraphs of three following types:

- a) maximal subgraphs of constant degree not intersecting h' ,
- b) connected components of the set A' ,
- c) subgraphs $\Delta_{\tilde{h}}$ from Definition 1

It is easy to show that each subgraph $\tilde{Q}$ of types a)-c) has the following properties.

Proposition 3. The subgraph $\tilde{Q}$ is incident to not more than two vertices of the graph $\tilde{L}_\infty + \tilde{g}_2$. The mapping F is not constant on these vertices. If $\tilde{Q}$ is incident to only one vertex $\tilde{a}$ of the graph $\tilde{L}_\infty + \tilde{g}_2$,

then

$$\det \tilde{Q} = \frac{n(\tilde{a})}{\text{Deg } \tilde{Q}} \det F(\tilde{Q}).$$

If $\tilde{Q}$ is incident to two vertices $\tilde{a}, \tilde{b} \in \tilde{L}_\infty + \tilde{g}_2$, then

$$\det \tilde{Q} = \frac{n(\tilde{a})n(\tilde{b})}{\text{Deg } \tilde{Q}} \det F(\tilde{Q}).$$

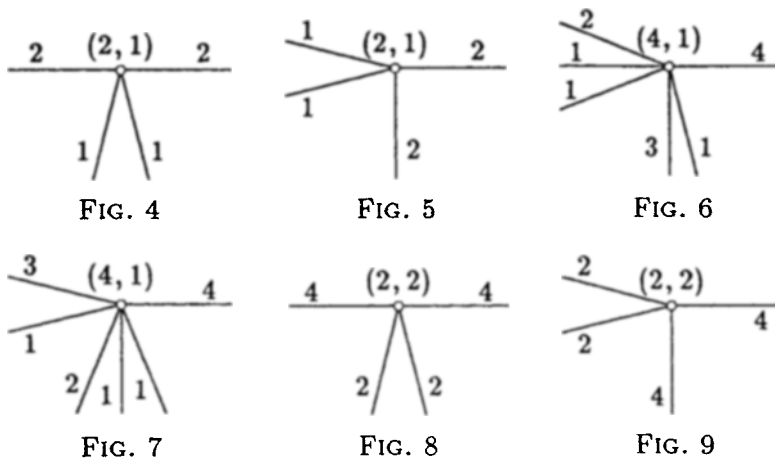
If $\tilde{a} \in \tilde{L}_\infty + \tilde{g}_2$ is a vertex incident to $\tilde{Q}$ and $\tilde{T} \subset \tilde{Q}$ is a linear chain incident to $\tilde{a}$, then $\text{Deg } \tilde{T} = \text{Deg } \tilde{Q}$.

(Observe that if there is only one dicritical component, then maximal subgraphs of constant degree satisfy Proposition 3). From Proposition 3 and [4, Proposition 1] it follows that if $\det A_h = 1$, then the connected components of the set A' have determinant 1 and do not contribute to the right-hand side of formula (5) in [4]. We shall omit them in the figures that follow.

Suppose a vertex $\tilde{a} \in \tilde{L}_\infty$ is contained in no subgraph of types a)–c) (consequently, $\tilde{a}$ is a fork). As in [4, §3], denote by $U_{\tilde{a}}$ the subgraph of $\tilde{L}_\infty$ containing $\tilde{a}$ and all subgraphs of types a)–c) incident to $\tilde{a}$. To each subgraph $U_{\tilde{a}}$ assign a diagram in the following way. The vertex of the diagram corresponds to $\tilde{a}$; write the pair of numbers $(m(\tilde{a}), n(\tilde{a}))$ at the vertex. Edges of the diagram correspond to connected components of the graph $(U_{\tilde{a}} \setminus \tilde{a})$; write the degrees of components on the corresponding edges. Edges to the left (respectively, to the right) of $\tilde{a}$ correspond to subgraphs whose images under F lie to the left (respectively, to the right) of a . The edges under $\tilde{a}$ correspond to subgraphs whose images under F lie over (or under) the vertex a (if $\det A_h = 1$, then we omit the connected component A' on the diagram).

When we described the subgraph $U_{\tilde{a}}$ in the case of one dicritical component [4], we used the fact that the determinants of its lower branches and the determinants of those left branches that are at the same time branches of the graph $\tilde{L}_\infty$ are pairwise coprime. In our case, one of the connected components $U_{\tilde{a}} \setminus \tilde{a}$ can be incident to $\tilde{g}_1$; therefore, all the determinants of the branches mentioned above, maybe except one, are pairwise coprime. Repeating *mutatis mutandis* the proof of Lemmas 8, 9 of the paper [4], we obtain

Proposition 4. The diagram corresponding to the subgraph $U_{\tilde{a}}$ is represented either in Figs. 10, 11, 13–16 of the paper [4] or in Figs. 4–9 here. In the case of Figs. 4, 7, 8, the vertex $\tilde{g}_1$ is incident to one of the lower branches of the graph $U_{\tilde{a}}$; in the case of Figs. 5, 6, it is incident to one of the left branches of the graph $U_{\tilde{a}}$; in the case of Figs. 6, 7, we have $\text{Deg}(\tilde{a}\tilde{g}_1) = 1$; in the case of Fig. 9, the vertex a is to the right of the vertex h .



As in [4, Corollary 5] we obtain

Proposition 5. Let us represent the subgraphs corresponding to Figs. 4, 5 by edges of degree 2; the subgraphs corresponding to Fig. 8, by edges of degree 4; the remaining subgraphs $U_{\tilde{a}}$, by their diagrams. Then the diagram corresponding to the graph $\tilde{L}_{\infty} + \tilde{g}_2$ is given either by Figs. 20–25 of the paper [4] or by Figs. 10–12 here.

3. Prohibition of diagrams corresponding to $\tilde{L}_{\infty} + \tilde{g}_2$. Suppose the diagram corresponding to the graph $\tilde{L}_{\infty} + \tilde{g}_2$ is given by one of the Figs. 10–12. Then, by Propositions 4, 5, the vertex $\tilde{g}_1$ is incident to a subgraph corresponding to a uniquely defined edge of the diagram. (We identify edges starting at the same vertex and corresponding to subgraphs of the same degree if these subgraphs are branches of $\tilde{L}_{\infty}$ and their images under the map F coincide.) Now suppose the diagram corresponding to $\tilde{L}_{\infty} + \tilde{g}_2$ is given by one of the Figs. 20–25 from [4]. Then there are several possibilities for the vertex $\tilde{g}_1$.

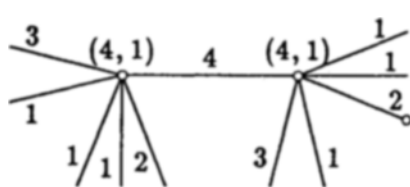


FIG. 10

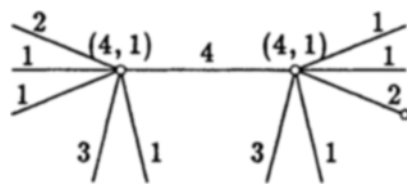


FIG. 11

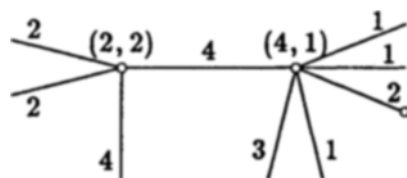


FIG. 12

Case 1. The vertex $\tilde{g}_1$ is incident to a subgraph $\tilde{Q}$ represented by a diagram edge. If the edge is “horizontal”, then either $\tilde{Q} = \Delta_{\tilde{h}}$ (see Definition 1) for some $\tilde{h} \in h'$ or $\text{Deg } \tilde{Q} = 2$ or, finally, $\text{Deg } \tilde{Q} = 4$ and $\tilde{Q}$ is represented by one of the diagrams given by Figs. 4, 6, 8. If the edge is “lower” than $\tilde{Q} \subset \tilde{A}'$, the graph L is given by Fig. 3, we have $\det A_h > 1$, and the vertex $\tilde{h} \in h'$ incident to $\tilde{Q}$ is a fork.

Case 2. The edge $\tilde{g}_1$ is incident to a subgraph $\tilde{Q}$ not represented on the diagram. Then Fig. 1 corresponds to the graph L , and the vertex $\tilde{h} \in h'$ incident to $\tilde{Q}$ is a fork.

For each diagram corresponding to $\tilde{L}_{\infty} + \tilde{g}_2$, we consider all possible positions of the vertex $\tilde{g}_1$ and prove their impossibility. In each case first we prove that the root of the graph $\tilde{L}$ lies in the subgraph $\langle \tilde{g}_1 \tilde{g}_2 \rangle$. Then, using formulas (3) from the paper [4] for the coefficients of the canonical divisor of the variety $\tilde{X}$ at $\tilde{g}_1$ and $\tilde{g}_2$, we obtain an equation and verify that our data do not satisfy it.

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